



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

**School of Computer Science
and Engineering**

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic x – Modeling of DC Motor control System

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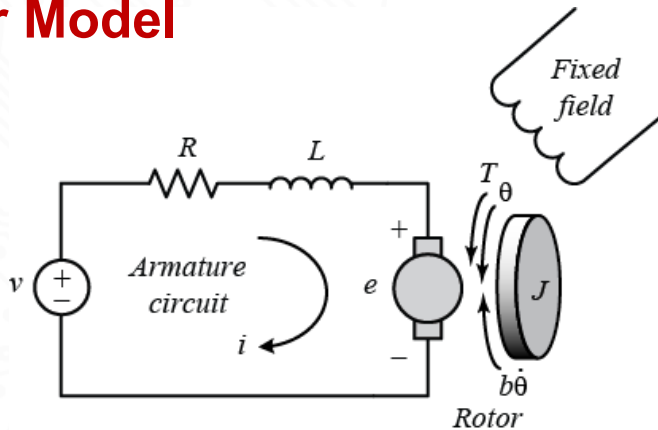
DC Motor Operation



When a DC voltage is applied to its terminal

- current flows through a coil of wire (i.e., inductor) inside the motor produces a magnetic field
- create a rotary motion

DC Motor Model



$$T = K_t i, \quad e = K_b \frac{d\theta}{dt}, \quad K_t = K_b = K$$

$$L \frac{di}{dt} + Ri = V - K \frac{d\theta}{dt} \quad \text{---- (1)}$$

$$J \frac{d^2\theta}{dt^2} + b \frac{d\theta}{dt} = Ki \quad \text{---- (2)}$$

DC Motor Model

$$L \frac{di}{dt} + Ri = V - K \frac{d\theta}{dt} \text{ ---- (1)}$$

$$J \frac{d^2\theta}{dt^2} + b \frac{d\theta}{dt} = Ki \text{ ---- (2)}$$

Relationship between:

Input = Voltage applied = V

Output = speed = $\frac{d\theta}{dt}$

$$(1) \Rightarrow (Ls + R)I(s) = V(s) - Ks\theta(s)$$

$$(2) \Rightarrow Js^2\theta(s) + bs\theta(s) = KI(s)$$

DC Motor Model in s Domain

Relationship between:

Input = Voltage applied = $V = V(s)$

Output = speed = $\frac{d\theta}{dt} = \dot{\theta}(s)$

$$(1) \Rightarrow (Ls + R)I(s) = V(s) - Ks\theta(s)$$

$$(2) \Rightarrow Js^2\theta(s) + bs\theta(s) = KI(s)$$

Transfer function of output speed against input voltage

$$\frac{\dot{\theta}(s)}{V(s)} = \frac{K}{(Js+b)(Ls+R)+K^2} \text{ rad/sec/V}$$

DC Motor Output Position

Transfer function of output speed against input voltage

$$\frac{\dot{\theta}(s)}{V(s)} = \frac{K}{(Js+b)(Ls+R)+K^2} \text{ rad/sec/V}$$

Transfer function of **output position** against input voltage

- **Integration** of speed (against time)

$$\begin{aligned} \frac{\theta(s)}{V(s)} &= \frac{1}{s} \frac{K}{(Js+b)(Ls+R)+K^2} \\ &= \frac{K}{s((Js+b)(Ls+R)+K^2)} \text{ rad/V} \end{aligned}$$

An example

$$J = 3.2284\text{E-}6;$$

$$b = 3.5077\text{E-}6;$$

$$K = 0.0274;$$

$$R = 4;$$

$$L = 2.75\text{E-}6;$$

$$\frac{\dot{\theta}(s)}{V(s)} = \frac{0.0274}{8.878\text{e-}^{12} s^2 + 1.291\text{e-}^5 s + 0.0007648}$$

Poles (for speed) are at $x = -59.24$
and $x = -1454100$

Position (Rotation Angle)

$$\frac{\theta(s)}{V(s)} = \frac{0.0274}{8.878e^{-12} s^3 + 1.291e^{-5} s^2 + 0.0007648s}$$

Poles (for position) are at $x = 0$,
 $x = -59.24$
and $x = -1454100$

Note that coefficient of s^3 is very small

- practically can be ignored

$$\frac{\theta(s)}{V(s)} = \frac{0.0274}{1.291e^{-5} s^2 + 0.0007648s}$$

Position (Rotation Angle)

$$\frac{\theta(s)}{V(s)} = \frac{0.0274}{1.291e^{-5}s^2 + 0.0007648s}$$

Poles (for position) are at $x = 0$,
 $x = -59.24$

That is

- the poles that was at $x = -1454100$ can be ignored.

Why?

Pole Positions and Time response

Comparing the transfer functions and their inverse Laplace transform:

$$\text{a) } \frac{Y(s)}{U(s)} = \frac{G_{\text{DC}}}{(1 + s\tau)} \quad \text{pole} = -\frac{1}{\tau} \quad \Rightarrow \quad y(t) = G_{\text{DC}}(1 - e^{-t/\tau})$$

$$\text{b) } \frac{Y(s)}{U(s)} = \frac{G_{\text{DC}}}{(1 - s\tau)} \quad \text{pole} = +\frac{1}{\tau} \quad \Rightarrow \quad y(t) = G_{\text{DC}}(-1 + e^{t/\tau})$$

When the **pole** is a **positive** value

- time domain expression will have a **+ve exponential** unbound term
- i.e. unstable system

Hence **pole** must be of **-ve** value for stable system

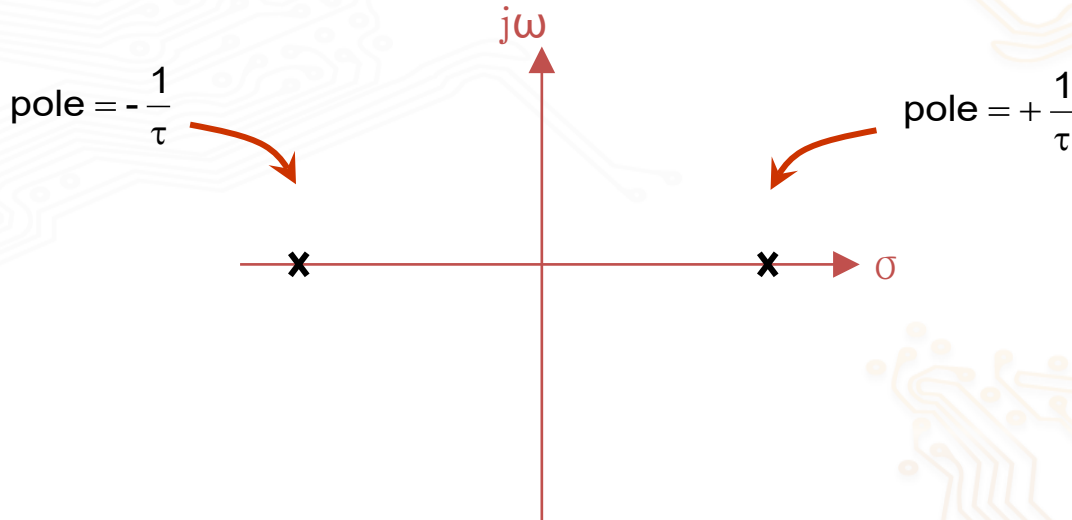
s-Plane

The poles (and zeros) of the s-domain transfer function

- are usually plotted on a two dimensions complex s-plane
- with real and imaginary axis

Location of the pole(s)

- indicated with a cross (x)



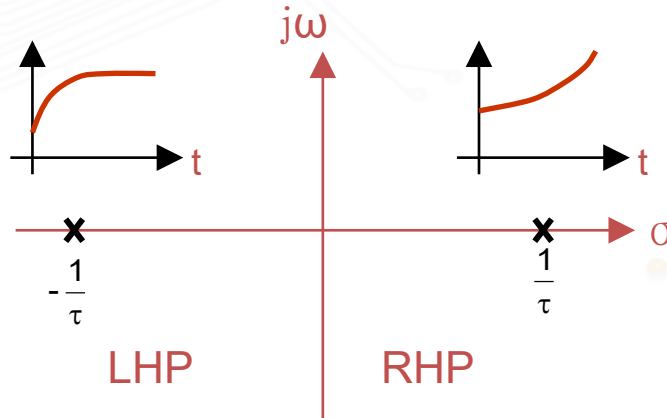
LHP & RHP

Since pole must have –ve real value for system to be stable

- **stable system** must have pole on the left side of the $j\omega$ axis i.e. the Left Half Plane (LHP)

Pole on right side of $j\omega$ axis will have +ve real value

- hence a system is considered **unstable** when it contains pole(s) on the Right Half Plane (RHP)



Pole and Time Response

Comparing the 1st order transfer function

$$\frac{Y(s)}{U(s)} = \frac{G_{DC}}{(1 + s\tau)} \quad \text{where pole} \equiv (1 + s\tau) = 0 \Rightarrow s = -\frac{1}{\tau}$$

with expression of step response:

$$y(t) = G_{DC}(1 - e^{-t/\tau})$$

Pole of the TF, **p** = **reciprocal** of system's time constant

- i.e. the bigger the magnitude of the pole
- the shorter is the time constant
- the faster the response of the system

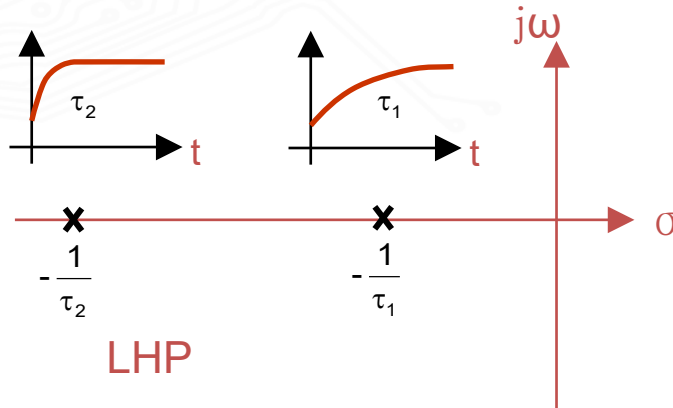
Response Time & Pole's Position

System with pole further to the left

- has faster response time

To improve the response of the system

- try 'pushing' the pole to further left half of the s-plane



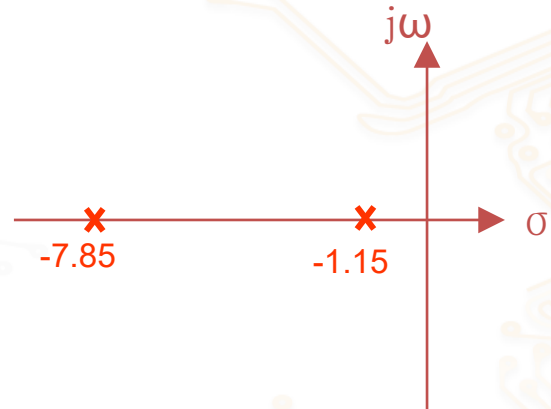
Example of 2nd Order System

Example 1: Transfer function of system

$$\frac{Y(s)}{U(s)} = \frac{9}{s^2 + 9s + 9}$$

Step response (i.e. $U(s) = 1/s$):

$$\begin{aligned} Y(s) &= U(s) \frac{9}{(s^2 + 9s + 9)} \\ &= \frac{1}{s} \frac{9}{(s^2 + 9s + 9)} \\ &= \frac{1}{s} \frac{9}{s(s + 1.15)(s + 7.85)} \\ &= \frac{1}{s} + \frac{0.17}{(s + 7.85)} - \frac{1.17}{(s + 1.15)} \end{aligned}$$



Poles: 2 real poles at -1.15 and -7.85

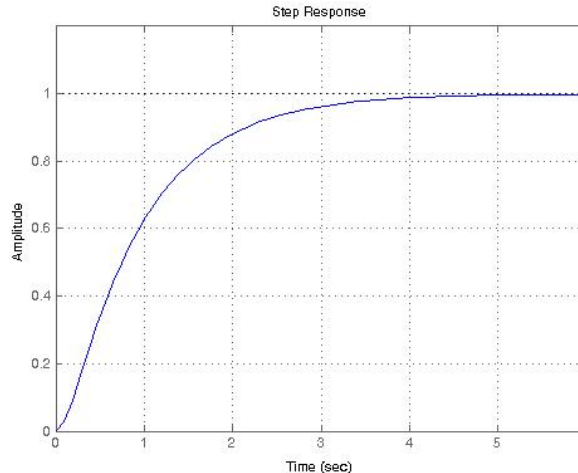
Example of 2nd Order System

Example 1 (cont)

$$\text{Step response } Y(s) : Y(s) = \frac{1}{s} + \frac{0.17}{(s+7.85)} - \frac{1.17}{(s+1.15)}$$

Inverse Laplace Transform: $y(t) = 1 + 0.17e^{-7.85t} - 1.17e^{-1.15t}$

i.e. an overdamped monotonic response, **dominated by the pole at -1.15**



Example of 2nd Order System

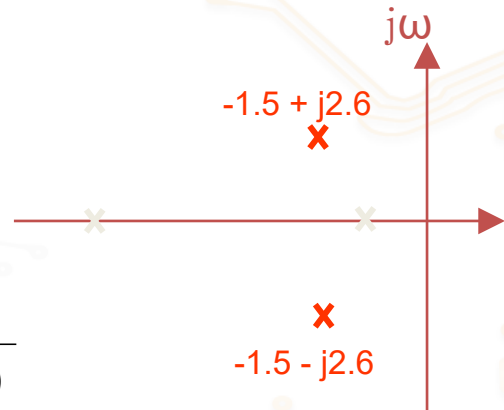
Example 2: Transfer function of system

$$\frac{Y(s)}{U(s)} = \frac{9}{s^2 + 3s + 9}$$

Step response:

$$\begin{aligned} Y(s) &= U(s) \frac{9}{(s^2 + 3s + 9)} \\ &= \frac{1}{s} \frac{9}{(s^2 + 3s + 9)} \\ &= \frac{1}{s} \frac{9}{(s + 1.5 + j2.6)(s + 1.5 - j2.6)} \end{aligned}$$

Poles: 2 complex conjugate poles
at $-1.50 \pm j2.60$



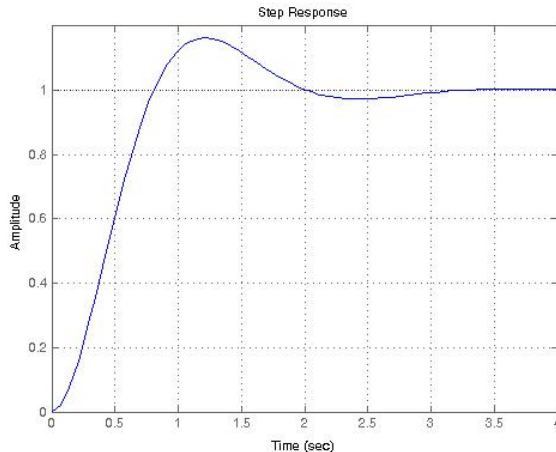
Example of 2nd Order System

Example 2 (cont)

Step response $Y(s)$:
$$Y(s) = \frac{1}{s} \frac{9}{(s + 1.50 + j2.60)(s + 1.50 - j2.60)}$$

Inverse Laplace Transform: $y(t) = 1 - 1.155e^{-1.50t} \cos(2.60t - 30^\circ)$

i.e. an underdamped oscillatory response



Note that the real part of the pole causes the exponential decay in the oscillation amplitude

Example of 2nd Order System

Example 3: Transfer function of system

$$\frac{Y(s)}{U(s)} = \frac{9}{s^2 + 9}$$

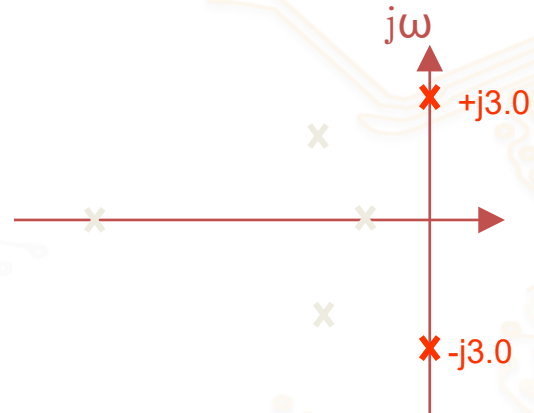
Step response:

$$Y(s) = U(s) \frac{9}{(s^2 + 9)}$$

$$= \frac{1}{s} \frac{9}{(s^2 + 9)}$$

$$= \frac{1}{s} \frac{9}{(s + j3)(s - j3)}$$

Poles: 2 imaginary poles at $\pm j3.0$



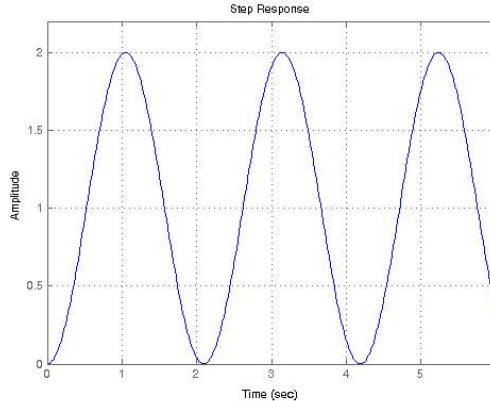
Example of 2nd Order System

Example 3 (cont)

Step response $Y(s)$:
$$Y(s) = \frac{1}{s} \frac{9}{(s + j3)(s - j3)}$$

Inverse Laplace Transform: $y(t) = 1 - \cos(3t)$

i.e. an oscillatory response (unstable)



Example of 2nd Order System

Example 4: Transfer function of system

$$\frac{Y(s)}{U(s)} = \frac{9}{s^2 + 6s + 9}$$

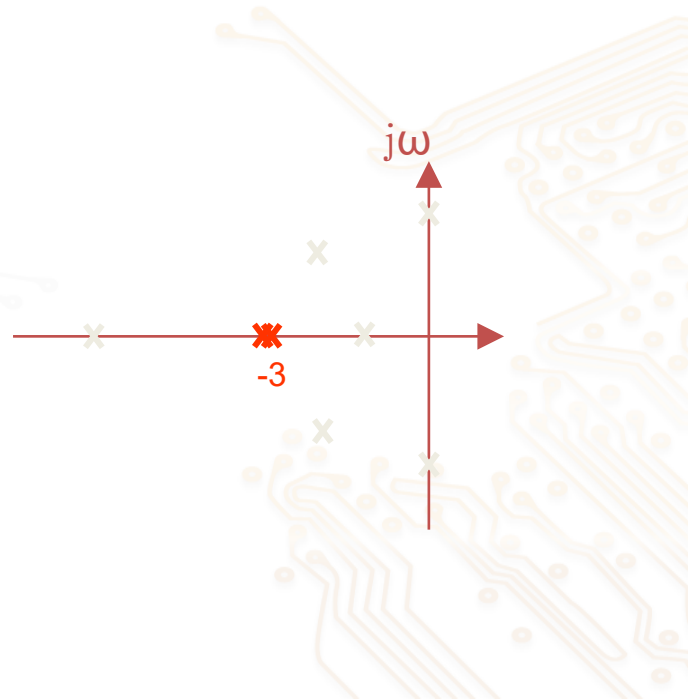
Step response:

$$Y(s) = U(s) \frac{9}{(s^2 + 6s + 9)}$$

$$= \frac{1}{s} \frac{9}{(s+3)(s+3)}$$

$$= \frac{1}{s} \frac{9}{(s+3)^2}$$

Poles: a double pole at $s = -3$



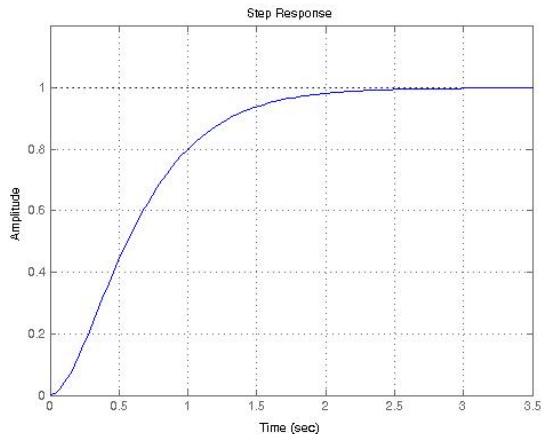
Example of 2nd Order System

Example 4 (cont)

Step response $Y(s)$:
$$Y(s) = \frac{1}{s} \frac{9}{(s+3)^2}$$

Inverse Laplace Transform:
$$y(t) = 1 - 3te^{-3t} - e^{-3t}$$

i.e. a critical damped response: Fastest response without overshoot
(because dominant pole is at the most -ve real possible value)



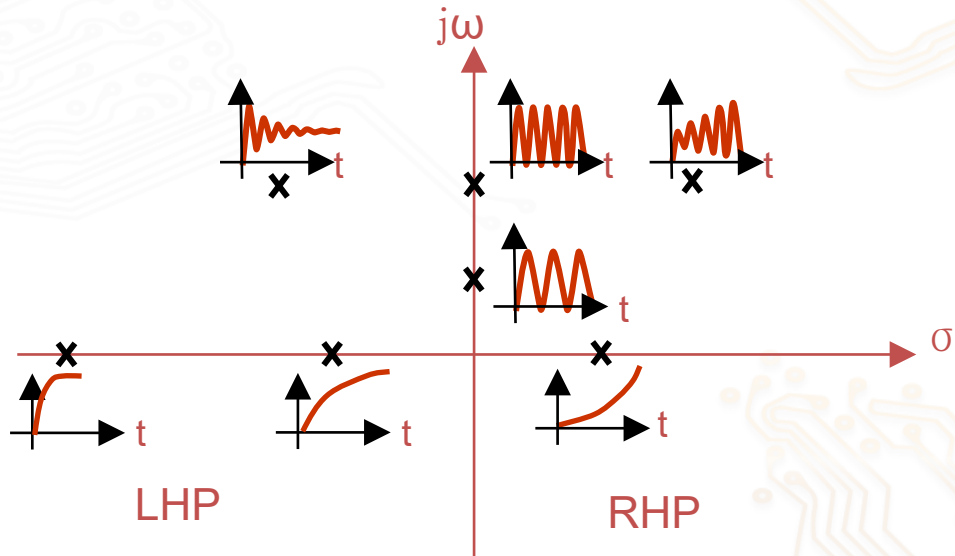
The transition point
between overdamped
and underdamped
response

System Response and Pole Position

Location of the (dominant) pole in the complex s-plane determines the type of response of the system

- LHP pole with real part only produces non-oscillatory response
- LHP complex poles produce oscillatory response

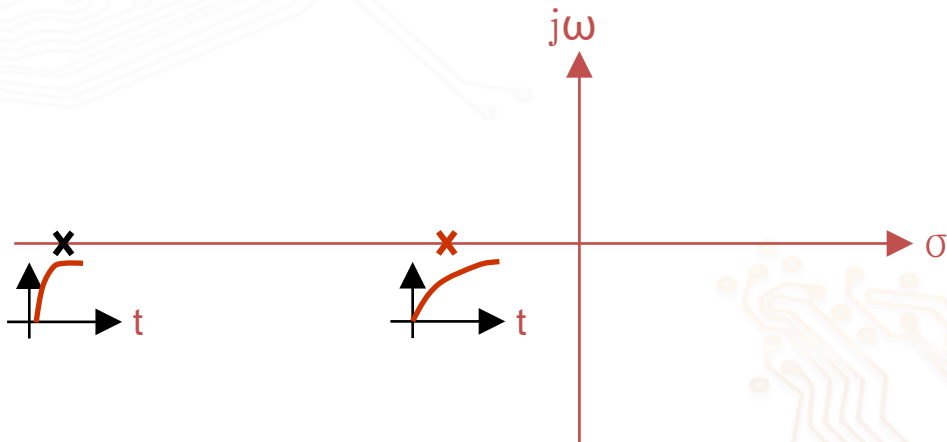
Any pole on the RHP will make the system unstable



Multiple Poles System

Systems that are 2nd order or higher will exhibit multiples poles in the s-plane

- response will be determined by the poles' positions
- typically **dominated by the one(s) that is(are) closer to the $j\omega$ axis (dominant poles)**
- especially when the others are more than 5 times further to the left



System with Zero

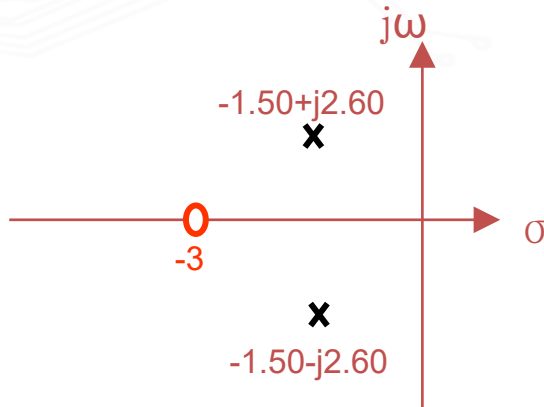
When a transfer function contains factor(s) of the form $(s+z)$ in its **numerator**

- $s = -z$ is referred as the **Zero** of the system

Example:

$$\frac{Y(s)}{U(s)} = \frac{9(s+3)}{(s+1.50+j2.60)(s+1.50-j2.60)}$$

The TF contains 2 complex poles and **one zero of -3**



Pole-Zero Cancellation

If a Zero is located very near/on top of a pole (e.g. by design)

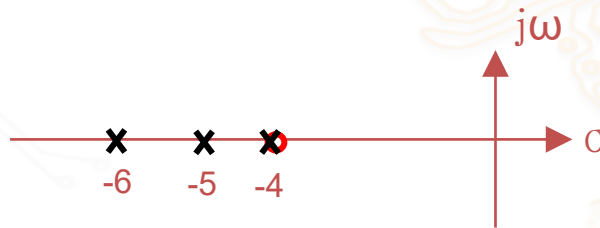
- the pole will effectively be cancelled by the Zero
- reduce the system order by 1
- response of the system will be determined by the other poles left in the system

Example:

$$Y(s) = \frac{1}{s} \frac{26.25(s + 3.99)}{(s + 4)(s + 5)(s + 6)}$$

$$Y(s) = \frac{0.87}{s} + \frac{0.033}{s + 4} - \frac{5.3}{s + 5} + \frac{4.4}{s + 6}$$

$$y(t) = 0.87 + 0.033e^{-4t} - 5.3e^{-5t} + 4.4e^{-6t}$$
$$\approx 0.87 - 5.3e^{-5t} + 4.4e^{-6t}$$



Small & negligible for pole at -4. (i.e. cancelled by the zero at -3.99)

Summary

Poles are equivalent to the time constants of a system

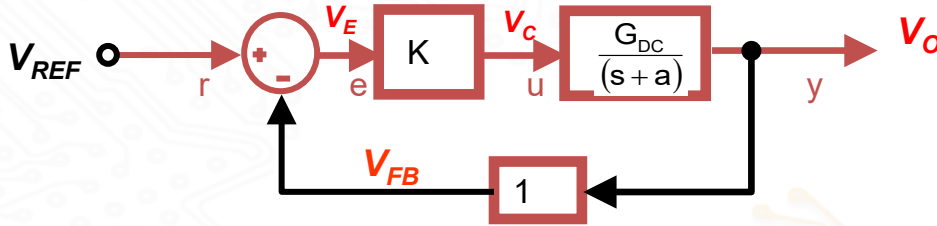
- hence similarly reveal the system response

Poles and Zero can be used to analyze the performance of the control loop & predict the time response of the system

- Poles must be on the LHP for the system to be stable
- Transient response of the system will be dominated by the pole(s) that are closer to the $j\omega$ axis.
- Complex pole will produce oscillatory transient response
- System with RHP pole will be unstable
- System with Zero will affect the response of the system

Hence a system should be designed, e.g. **by using feedback**, such that **poles are placed further to the left for faster response**.

A Proportional Control System



$$\frac{V_O}{V_{REF}} = \frac{G}{(1+GH)} \quad \text{where} \quad G = K \frac{G_{DC}}{(s+a)} \quad \text{and} \quad H=1$$

$$\frac{V_O}{V_{REF}} = \frac{KG_{DC} \cancel{(s+a)}}{\left(1 + \frac{KG_{DC} \cancel{(s+a)}}{(s+a)}\right)} = \frac{KG_{DC}}{(s+a) + KG_{DC}} = \frac{KG_{DC}}{s + (a + KG_{DC})}$$

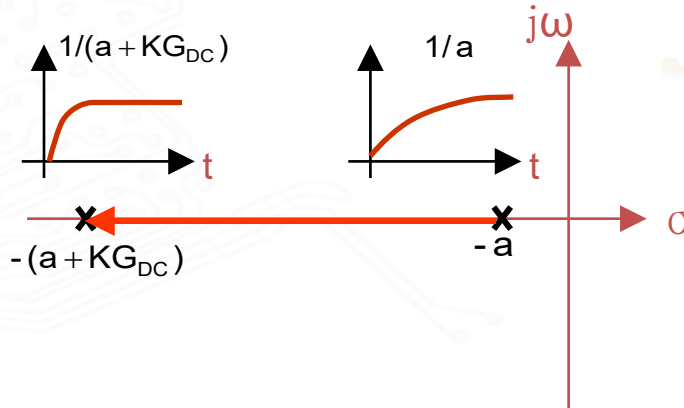
The pole of the close loop TF is now moved to $s = -(a+KG_{DC})$

- i.e. much further to the left in the s-plane
- response time will improve with negative feedback

Response Time & Pole's Position

Negative feedback improves the response time of the system

- by 'pushing' the pole further to the left of the s-plane
- achieved by increasing the value of K

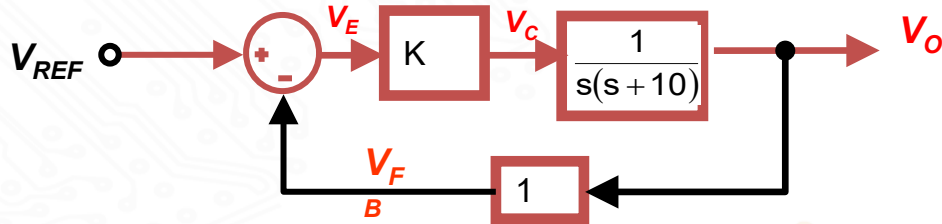


Example: if $a = 10$ (i.e. $\tau = 100$ msec), $G = 1$

With $K = 1000$,

- New pole position = -1010 , $\Rightarrow \tau = 1/1010 = 1$ msec
- i.e. 100 time improvement in step response

2nd order Close Loop System



Example: If $a = 10$ and $G_{DC} = 1$

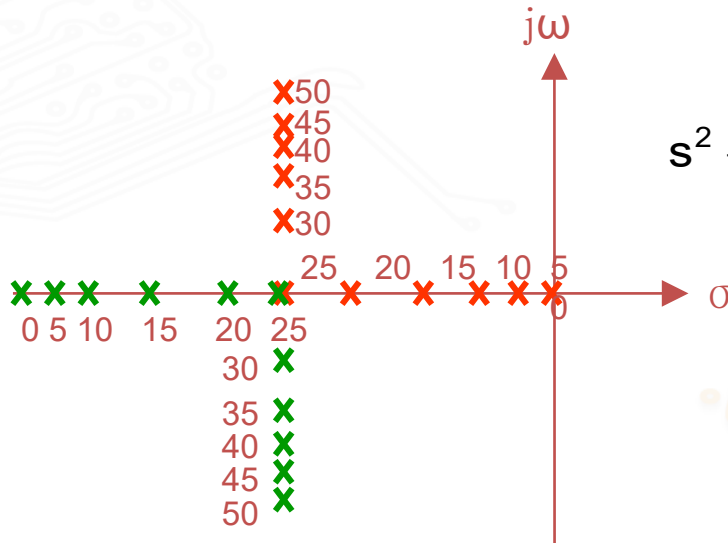
Pole of the system is solution of the characteristics equation:

$$s^2 + 10s + K = 0$$

K	0	5	10	15	20	25	30	35	40	45	50
P_1	0	-0.53	-1.13	-1.84	-2.76	-5	$\frac{-5}{+j2.24}$	$\frac{-5}{+j3.16}$	$\frac{-5}{+j3.87}$	$\frac{-5}{+j4.47}$	$\frac{-5}{+j5}$
P_2	-10	-9.47	-8.87	-8.16	-7.24	-5	$\frac{-5}{-j2.24}$	$\frac{-5}{-j3.16}$	$\frac{-5}{-j3.87}$	$\frac{-5}{-j4.47}$	$\frac{-5}{-j5}$

2nd order Close Loop System

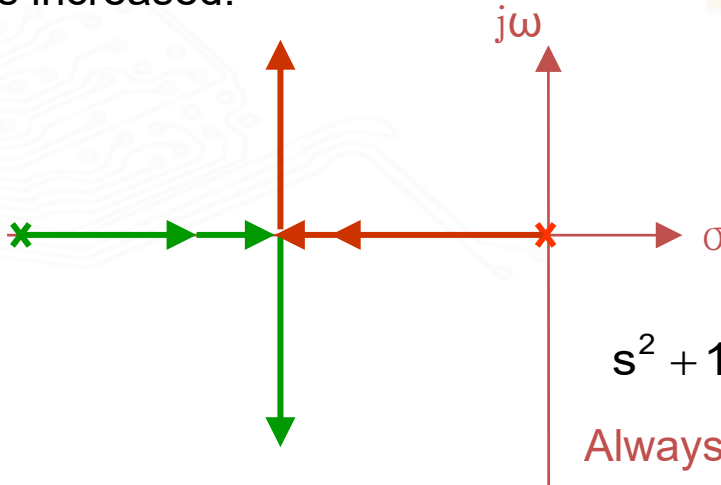
K	0	5	10	15	20	25	30	35	40	45	50
P_1	0	-0.53	-1.13	-1.84	-2.76	-5	-5 +j2.24	-5 +j3.16	-5 +j3.87	-5 +j4.47	-5 +j5
P_2	-10	-9.47	-8.87	-8.16	-7.24	-5	-5 -j2.24	-5 -j3.16	-5 -j3.87	-5 -j4.47	-5 -j5



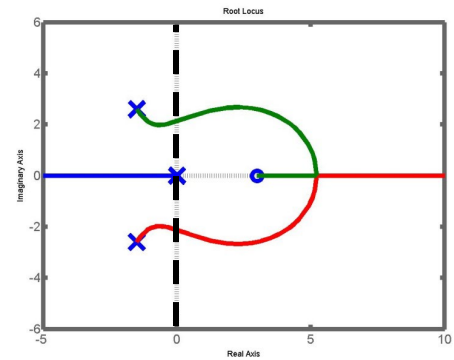
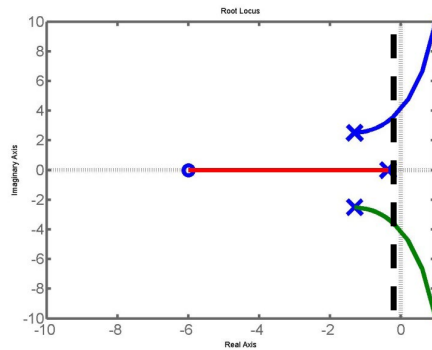
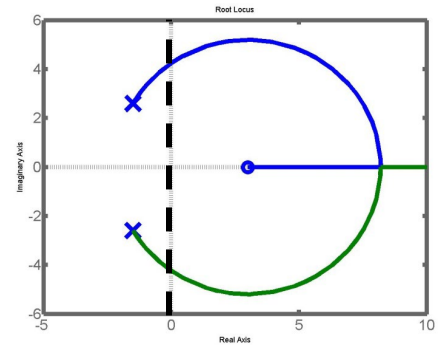
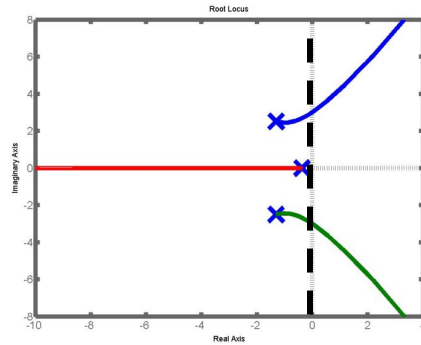
Aside: Root Locus

Root Locus: **Path taken** by the movement of the closed loop system poles as the controller gain K is increased

- provides information on how the dynamic response will be like as K is increased.



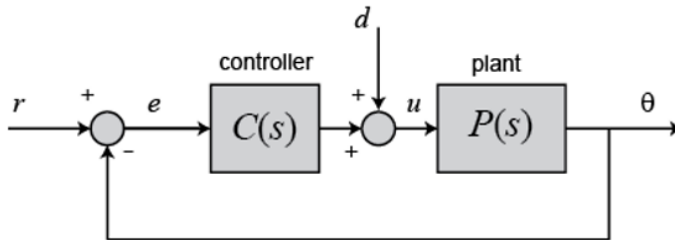
Examples of Root Locus Plot



PID based Control System

By adding a PID controller into a system

- we are introducing more Poles and Zeros
- which we can use to adjust the system response.



PD controller: $G_c(s) = K_p + K_d s$

PI controller: $G_c(s) = K_p + K_i \frac{1}{s} = \frac{K_p s + K_i}{s}$

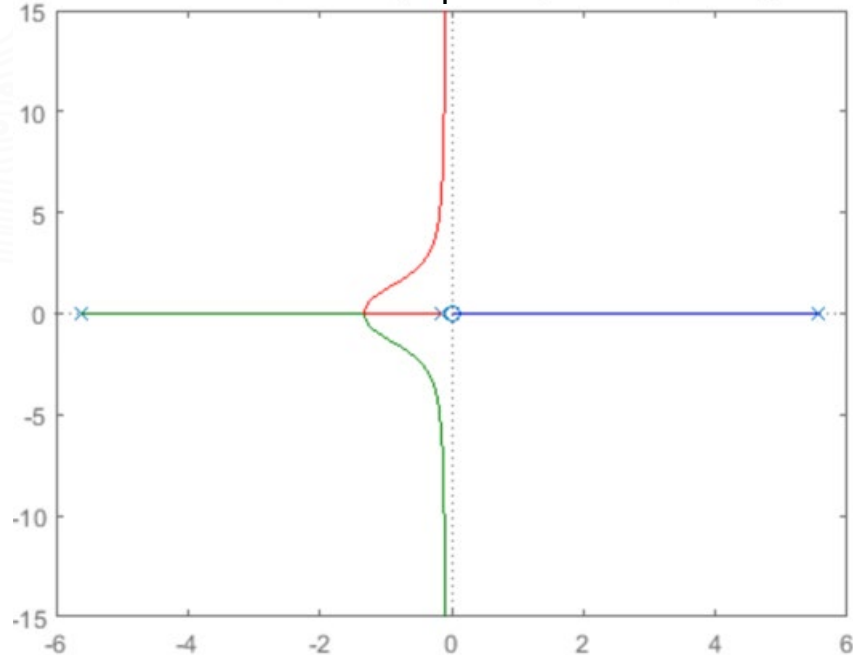
PID controller: $G_c(s) = K_p + K_d s + K_i \frac{1}{s} = \frac{K_i + K_p s + K_d s^2}{s}$

Poles and Zero of a Self-Balancing Robot

Zero: 0

Poles: 5.5652, -5.6041, -0.1428

Root locus with Proportional Control

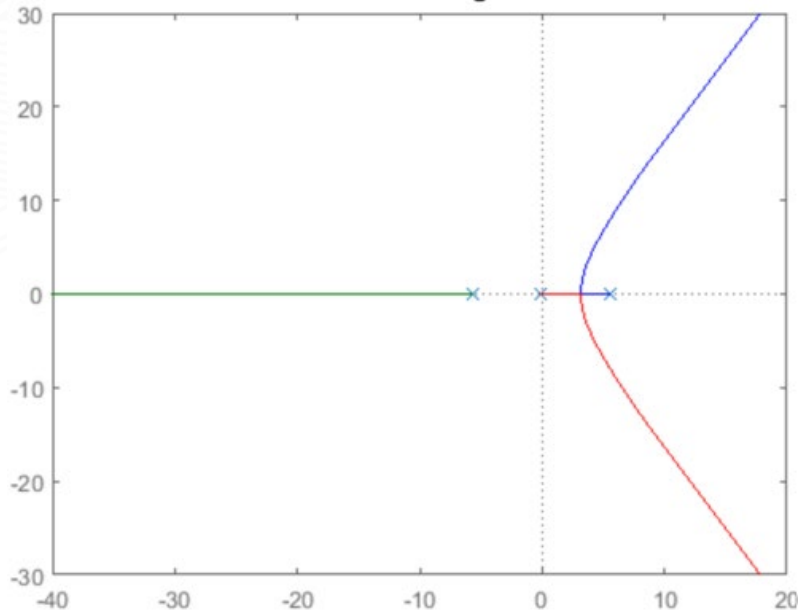


PI Control of a Self-Balancing Robot

Add the I with pole at origin to cancel the Zero

- Poles: 5.5652, -5.6041, -0.1428

Root locus with PI Control

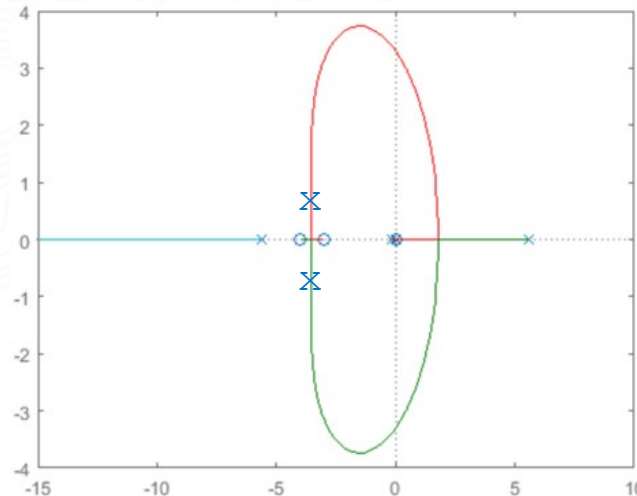


PID control of a Self-Balancing Robot

Add D to introduce Zero(s) on the LHP

- to 'pull' the RHP root local branches toward the LHP
- choose K with dominant closed-loop poles in the LHP

Root locus with PID Control



Transient Response

